Passermark – Command line

Manual for `passermark-cli` with case studies · Passermark 1.12.6

The command line runs Passermark's heavy functions without a GUI: in scripts, for whole folders at once, or later automatically in the background. Everything described here does the same as the corresponding dialogs in the program.

1 · Invocation

```
passermark-cli <job> <input.pdf> <output.pdf> [options]
```

System	What the command is called
Linux (installed with install.sh)	`passermark-cli …`
Linux (AppImage)	`./Passermark-*-x86_64.AppImage --cli …`
Windows	`"C:\Program Files\Passermark\passermark-cli.exe" …`
Windows (path option ticked during installation)	`passermark-cli …`

Windows: since 1.7 the setup has the option “Make passermark-cli available in every command prompt”. Then `passermark-cli` is enough – in a **newly opened** Command Prompt or PowerShell window. Without the option, always give the full path in quotation marks.

The input file is never modified. The output is only written when everything is finished – after an error or cancellation no half-written file is left behind. Input and output must not be the same file.

The nine jobs

Job	What it does
`cutcontour`	Cut line for a cutting plotter (spot colour CutContour), optionally with bleed
`separate`	Detect motifs, each motif as its own page
`manip`	CMYK conversion with an ICC profile and/or cropping to a format
`repair`	Repair, optimise for printing, shrink, PDF/A, password
`preflight_fix`	Flatten layers, embed fonts, outline text, flatten transparency
`impose`	Imposition as in the print dialog: booklets and signatures, step & repeat, N-up, posters
`vdp`	Variable data: numbers, texts from CSV, QR codes, Code 128, EAN-13 – every page the next record
`split`	Split pages: halves or a grid, e.g. a booklet exported as spreads into single pages
`shrink`	Reduce file size: lossless measures can be switched individually, optional image downsampling

Options

Option	Meaning
`--set name=value`	Set one setting; can be repeated. Numbers, `true`/`false` and lists are recognised, everything else is text. Nested with a dot: `--set detect.tolerance=40`
`--preset file.json`	Load settings from a file (see section 3). `--set` overrides individual values from it
`--preset "Name"`	Use a preset saved in the program (see section 3)
`--pages 1,3-5`	Process only these pages (where applicable)
`--quiet`	No progress display
`--json-progress`	Progress and result as JSON lines – for other programs

Option	Meaning
`list`	Show all jobs: `passermark-cli list`
`settings <job>`	All settings of a job with their defaults as JSON – the template for your own presets
`presets [job]`	List the presets saved in the program
`beispiele [folder]`	Copy the variable-data examples to the user folder (default: `Passermark-Beispiele`) and register them as presets “Beispiel 1 …”
`datenarten`	Data kinds for tables (business card, Wi-Fi, e-mail, Code 128, EAN-13 …) with their columns
`datenvorlage <kind> [file.csv]`	Empty table with the matching columns and one example row
`datencheck <file.csv> [kind]`	Check a table – reports row and error; return code 1 on problems

Return codes and cancelling

`0` = done · 1 = error (e.g. file missing, no motif found) · 2 = wrong invocation · `130` = cancelled. **Ctrl+C** cancels cleanly, without an output file. In scripts you can react to errors this way (section 4).

2 · The jobs with case studies

cutcontour – cut line for the plotter

Setting	Default	Meaning
`shape`	`contour`	Shape: `contour` (follows the motif), `rect`, `rounded`, `circle`, `oval`, `hexagon`, `octagon`, `heart`, `star`, `shield`, `arch`
`offset_mm`	0	Distance of the line from the motif: + outwards (white border), − inwards
`smooth_mm`	1,5	Smoothing of the contour (deliberately does not follow the motif exactly)
`bleed`	`true`	Create bleed (colour runs beyond the cut line)
`bleed_mm`	2	Width of the bleed
`inner`	`false`	Also cut inner contours (holes in the motif)
`width_mm` / `height_mm`	0	Basic shape: size of the cut line in mm (0 = from the motif)
`shift_x_mm` / `shift_y_mm`	0	Basic shape: offset from the motif centre (+ right / + up)
`corner_mm`	3	Corner radius for `rounded`
`single_shape`	`true`	Basic shape: one shape around the whole motif; `false` = one per object (sticker sheet)
`per_object`	`false`	Each object on its own page
`margin_mm`	6	Page margin for `per_object`
`spot`	`CutContour`	Name of the spot colour – differs depending on the RIP software
`stroke_pt`	0,25	Line width
`dpi`	200	Accuracy; 300 easily takes twice as long
`detect.*`		Motif detection, see `separate`

Sticker sheet with contour and 2 mm bleed

Every sticker gets its contour, the colour runs 2 mm beyond the line – no white flash when cutting.

```
passermark-cli cutcontour bogen.pdf bogen-cut.pdf --set bleed_mm=2
```

Logo with an 80 × 50 mm rectangle, without bleed

Fixed size of the cut line, centred on the whole logo – even if the logo consists of several parts.

```
passermark-cli cutcontour logo.pdf logo-cut.pdf \
    --set shape=rect --set width_mm=80 --set height_mm=50 --set bleed=false
```

Round stickers Ø 60 mm on a sheet

Every sticker individually (`single_shape=false`), each circle exactly 60 mm.

```
passermark-cli cutcontour bogen.pdf rund.pdf \
    --set shape=circle --set width_mm=60 --set height_mm=60 --set single_shape=false
```

3 mm white border around the motif

The contour lies 3 mm outside; without bleed the border stays white.

```
passermark-cli cutcontour logo.pdf logo-rand.pdf --set offset_mm=3 --set bleed=false
```

Every sticker on its own page

Practical for individual jobs or when the plotter cuts page by page.

```
passermark-cli cutcontour bogen.pdf einzeln.pdf --set per_object=true --set margin_mm=5
```

Different spot colour for the RIP software

Some programs expect a different name than “CutContour”, e.g. “Thru-cut”.

```
passermark-cli cutcontour bogen.pdf bogen-cut.pdf --set spot=Thru-cut
```

Only specific pages

Process pages 1 and 3 to 5.

```
passermark-cli cutcontour mappe.pdf mappe-cut.pdf --pages 1,3-5
```

separate – motifs as individual pages

Setting	Default	Meaning
`margin_mm`	0	Margin around each motif (negative = inwards)
`detect.mode`	`auto`	`auto`, `transparent` (cut out) or `color` (solid-colour background)
`detect.tolerance`	28	How much the motif must differ from the background (0–255); higher = less sensitive to spots
`detect.min_size_mm`	5	Smaller parts count as dust
`detect.gap_mm`	1	Parts lying closer together belong to one motif

Split a scan with several motifs

Each motif becomes a page, with a 3 mm margin.

```
passermark-cli separate scan.pdf motive.pdf --set margin_mm=3
```

Blotchy background (e.g. recycled paper)

Higher tolerance and larger minimum size – spots are no longer detected as motifs.

```
passermark-cli separate scan.pdf motive.pdf \
    --set detect.tolerance=40 --set detect.min_size_mm=10
```

manip – CMYK conversion and cropping

Setting	Default	Meaning
`crop`	`false`	Bring to a format (excess cut equally on both sides)
`crop_scale`	`true`	First scale until the format is completely filled (A4 → A6 = 50 %), then trim the excess of the other edge; `false` = only trim (files with bleed)
`crop_inset_mm`	0	Trim this much all around beforehand (white borders of the original)
`crop_size`	A4	A0–A7, B4, B5, C4–C6, DL, SRA3, A3+, Letter, Legal, Tabloid, “business card 85×55”, “business card 90×50”, “square 210”, “square 148” or `custom`
`crop_w_mm` / `crop_h_mm`	210 / 297	Dimensions for `custom`
`crop_follow`	`true`	Follow the page's portrait/landscape orientation
`cmyk`	`false`	Convert to CMYK (needs Ghostscript)
`target`		Path to the CMYK destination profile (.icc), e.g. ISOcoated_v2_eci.icc
`cmyk_mode`	`rgb_only`	`rgb_only` (existing CMYK stays), `all` (also convert CMYK, source profile in `source`), `gray`
`intent`	`relative`	`perceptual`, `relative`, `saturation`, `absolute`
`gray_to_k`	`true`	Print grey/black with K only

Bring all pages to A5

A4 is scaled down to 70.7 %; a narrower motif fills the width, an equal amount is trimmed top and bottom.

```
passermark-cli manip flyer.pdf flyer-a5.pdf --set crop=true --set crop_size=A5
```

File with 3 mm bleed to trim size (trim only)

```
passermark-cli manip flyer-anschnitt.pdf flyer-a4.pdf \
    --set crop=true --set crop_size=A4 --set crop_scale=false
```

Custom format 100 × 150 mm

```
passermark-cli manip foto.pdf foto-10x15.pdf \
    --set crop=true --set crop_size=custom --set crop_w_mm=100 --set crop_h_mm=150
```

Print data to CMYK (ISO Coated v2)

The profile is available free of charge from the ECI (eci.org). Adapt the path to where you store it.

```
passermark-cli manip druck.pdf druck-cmyk.pdf \
    --set cmyk=true --set target=/pfad/ISOcoated_v2_eci.icc
```

repair – repair, optimise, PDF/A, password

Setting	Default	Meaning
`mode`	`print`	`repair` (repair only), `print` (for sharing and printing, embed fonts), `screen` (shrink for e-mail, 150 dpi), `pdfa` (PDF/A-2b, archive/authorities)
`password`		Password of the input file if it is protected
`new_password`		New password for the output (empty = none); not with `pdfa` – PDF/A forbids encryption, the password is then omitted
`allow_print`	`true`	With a password: still allow printing

Prepare a broken PDF for printing

```
passermark-cli repair kaputt.pdf ok.pdf --set mode=print
```

Shrink for e-mail

```
passermark-cli repair katalog.pdf katalog-klein.pdf --set mode=screen
```

Archive as PDF/A

For archives, authorities and e-invoice attachments.

```
passermark-cli repair vertrag.pdf vertrag-a.pdf --set mode=pdfa
```

Protect with a password

Opening only with a password, printing stays allowed.

```
passermark-cli repair angebot.pdf angebot-geschuetzt.pdf --set new_password=geheim
```

preflight_fix – make problem PDFs print-ready

Setting	Default	Meaning
`fix`	`flatten_layers`	`flatten_layers` (fix layers – exactly what is visible is printed), `embed_fonts`, `outline_text` (text to paths), `flatten_transparency`, `flatten_softmask` (transparent gradients to opaque images – against black areas on the RIP, no Ghostscript needed)
`font_map`	`{}`	With `embed_fonts`: substitute fonts {font name: path}

Setting	Default	Meaning
dpi	300	With flatten_transparency and flatten_softmask: resolution of the rasterised areas

Plan with layers: print only the visible

```
passermark-cli preflight_fix plan.pdf plan-fest.pdf --set fix=flatten_layers
```

Old RIP: flatten transparency

```
passermark-cli preflight_fix plakat.pdf plakat-flach.pdf \
    --set fix=flatten_transparency --set dpi=300
```

embed_fonts, outline_text, flatten_transparency, repair and the CMYK conversion need Ghostscript. The Windows setup and AppImage include it; with the Linux installation install.sh installs it.

shrink – reduce file size

Every measure is a separate switch – if one causes problems, just turn it off with --set name=false. No Ghostscript needed. If the result is not smaller, the file is taken over unchanged.

Setting	Default	Meaning
rewrite	true	rewrite: old saved versions removed, compressed object streams, stronger ZIP
unreferenced	true	remove unused resources
dedupe	true	merge images, fonts, ICC profiles stored twice
private	true	remove Illustrator/Photoshop editing data (PieceInfo)
thumbs	true	remove embedded page thumbnails
pack	true	compress images losslessly (ZIP with predictor), remove ZIP around JPEG – only if smaller
bits16	false	16-bit images to 8 bit
attachments	false	remove embedded files (attachments)
hidden	false	remove content of hidden layers
downsample	false	downsample images above target_dpi × threshold (measured at the largest placement) to target_dpi
target_dpi	300	target resolution for downsampling
threshold	1.5	only above target × threshold (1.5 at 300 dpi = from 450 dpi)
photos_jpeg	false	save photos stored as ZIP as JPEG (real photos only, only if clearly smaller)
jpeg_quality	92	JPEG quality for downsampling and for photos_jpeg

Reduce losslessly only

The printed result stays pixel-identical.

```
passermark-cli shrink flyer.pdf flyer-klein.pdf
```

Poster: images to 150 dpi

Keep the editing data.

```
passermark-cli shrink plakat.pdf plakat-klein.pdf \
    --set downsample=true --set target_dpi=150 --set private=false
```

impose – impose booklets, signatures and step & repeat

The same as the print dialog, but as a PDF file: every output page is a press sheet. The sheet format is set in `sheet`. `passermark-cli settings impose` shows all print dialog settings; the most important:

Setting	Default	Meaning
`sheet`	`A4`	Sheet: A0–A6, A3+, SRA3, B4–B6, LETTER, LEGAL, TABLOID or `custom` with `sheet_w_mm`/`sheet_h_mm`
`handling`	`size`	`booklet`, `multiple` (several per sheet), `poster`
`booklet_kind`	`saddle`	`saddle` = saddle stitch (all sheets nested), `grouped` = signatures of several sheets, `stack` = single sheets (perfect binding)
`booklet_per_sig`	4	With `grouped`: sheets per signature (4 sheets = 16 pages)
`booklet_creep_mm`	0	Creep per sheet, roughly the paper thickness (0.1 for 80 g/m²)
`booklet_blanks`	`end`	Blank pages `end` or `before_back` (the back cover stays the last page)
`booklet_blank_text`	–	Notice text on added blank pages
`booklet_fold_marks`	`false`	Fold marks
`booklet_reg_marks`	`false`	Registration marks
`booklet_collation_marks`	`false`	Collation marks on the spine of each signature
`booklet_gutter_mm`	0	Gutter (distance to the fold)
`step_repeat`	`false`	Step & repeat: one page several times per sheet; `sr_by=long`/`short` + `sr_mm` for the repeat size
`bleed_mm`, `crop_marks`	0, `false`	Bleed (mirrored) and crop marks

A5 booklet on A4 (saddle stitch)

Blank pages before the back cover, fold marks.

```
passermark-cli impose heft.pdf heft-bogen.pdf --set handling=booklet --set sheet=A4 \
    --set booklet_blanks=before_back --set booklet_fold_marks=true
```

Thick book in 16-page signatures

4 sheets per signature, creep 0.1 mm, collation marks.

```
passermark-cli impose buch.pdf buch-bogen.pdf --set handling=booklet --set sheet=A3 \
    --set booklet_kind=grouped --set booklet_per_sig=4 --set booklet_creep_mm=0.1 \
    --set booklet_collation_marks=true
```

A4 flyer as A6 8× on SRA3

Long edge 148 mm, 2 mm bleed, crop marks.

```
passermark-cli impose flyer.pdf flyer-sra3.pdf --pages 1 --set sheet=SRA3 \
    --set step_repeat=true --set sr_by=long --set sr_mm=148 \
    --set bleed_mm=2 --set crop_marks=true
```

Print presets from the program apply here directly – including the paper size: set them in the print dialog, **Save…** at the top, then `--preset "Name"`. The print order of the sheets is designed for duplex printing over the **short edge**. If the PDF has a trim size with bleed (TrimBox, e.g. from InDesign with “use bleed”), the trim size is placed and the bleed comes from the document – `sr_mm` then refers to the trim size.

vdp – variable data: numbers, names, QR and bar codes

Places fields on the template – every page gets the next record. Records come from a CSV file (first line = column names; the separator ; , or tab is detected) or from pure numbering (`count`). Contents are templates: `{{Spalte}}`, `{{nr}}` (formatted number), `{{i}}` (running from 1). Everything stays vector.

The easiest way to create the fields is in the program (Variable data workspace, Ctrl+Shift+D, place the fields with the mouse, preview per record) and save a preset. The command line then takes the preset and changes only what differs per job:

Setting	Default	Meaning
`csv_path`	–	CSV file; empty = numbering only
`count`	1	Without CSV: this many copies
`records`	–	Only these records, e.g. 1-50
`numbering.start` / `.step` / `.digits`	1 / 1 / 0	Start, step, zero-padded to this many digits
`numbering.prefix` / `.suffix`	–	Text before/after the number
`numbering.check`	`none`	`luhn`, `ean`, `mod11` – append a check digit
`numbering.continue_key`	–	Counter name: the next job continues counting there
`reverse`	`false`	Backwards (tear-off pad: top sheet has the highest number)
`pages_per_record`	1	With each: this many pages per record – 2 = front and back
`fields[].pages`	–	Field only on these template pages: `1,3-4`, `ungerade`, `gerade`, `ungerade 1-50`
`order`	each	each = every page the next record (one-page template: one page per record), `record` = all pages per record, page = like record, page by page
`log_path`	–	Code log as CSV (output page, record, values)
`placeholders`	`false`	Take {{…}} in the PDF as the field position, remove the placeholder text
`fields[].qr_type`	`text`	QR field: `vcard`, `wifi`, `email`, `url`, `phone`, `sms`, `event`, `geo` – content from the table columns (in the preset; in the program “QR code type”)
`fields[].qr_map`	`{}`	Mapping of item → CSV column, e.g. `{"Position": "Title"}`; `-` = leave empty; items not mentioned are recognised automatically (also Outlook/Google exports)

Ready-made examples with template, CSV and presets (QR code, Code 128, EAN-13) are fetched by `passermark-cli beispiele`; explained in detail in the “Variable data” manual (Help → Variable data – Manual).

500 numbered tickets

Create the "Tickets" preset in the program first (field with {{nr}}).

```
passermark-cli vdp ticket.pdf tickets.pdf --preset "Tickets" --set count=500
```

Name badges from a guest list

Fields {{Name}} and a QR code with {{Name}}; plus a log file.

```
passermark-cli vdp schild.pdf schilder.pdf --preset "Namensschilder" \
    --set csv_path=gaeste.csv --set log_path=codes.csv
```

Business cards with a QR code from a contact list

Create an empty table, fill it in Excel, check, generate. The preset has a QR field of type “business card” (qr_type vcard).

```
passermark-cli datenvorlage vcard kontakte.csv
passermark-cli datencheck kontakte.csv
passermark-cli vdp karte.pdf karten.pdf --preset "Visitenkarte" --set csv_path=kontakte.csv
```

Every card with its own code 8× on SRA3, cut and stack

First the data, then imposition – the next page per repeat, the stacks stay in order.

```
passermark-cli impose tickets.pdf tickets-sra3.pdf --set sheet=SRA3 \
    --set step_repeat=true --set sr_sequence=true --set sr_stack=stack \
    --set crop_marks=true
```

split – split pages (halves, grid, spreads)

Splits pages into single pages – lossless, the content is not modified. If the page has a trim size, it is split within it and the bleed stays outside. Rotated pages are split the way you see them. In the program: select pages, right-click → Split → halve; or Pages → Split pages … / Split spreads.

Setting	Default	Meaning
`mode`	`halves_v`	`halves_v` = left | right, `halves_h` = top / bottom, `grid` = grid
`cols` / `rows`	2 / 2	Only with `grid`: columns × rows
`only_spreads`	`false`	Only split pages that are twice as wide (or grid-times as large) as the others
`rtl`	`false`	Order right to left (right binding)
`use_trim`	`true`	Split within the trim size, keep the bleed outside

Booklet exported as spreads

Page 1 single A4, then A3 spreads: halve only the spreads – gives consecutive A4 pages.

```
passermark-cli split broschuere.pdf einzelseiten.pdf --set only_spreads=true
```

Sheet with 2 × 4 cards into single cards

Grid on page 1.

```
passermark-cli split bogen.pdf karten.pdf --set mode=grid --set cols=2 --set rows=4 --pages 1
```

3 · Presets: settings as a file

Anyone who needs the same settings again and again stores them once in a file. The template with all settings and their defaults is provided by `settings`.

```
passermark-cli settings cutcontour > sticker.json
```

Then adapt the file in a text editor – only change the values after the colon:

```
{
  "job": "cutcontour",
  "settings": {
    "shape": "rounded",
    "corner_mm": 4,
    "offset_mm": 2,
    "bleed_mm": 2,
    ...
  }
}
```

Use it – individual values can still be changed with `--set`:

```
passermark-cli cutcontour bogen.pdf bogen-cut.pdf --preset sticker.json
passermark-cli cutcontour bogen.pdf bogen-cut.pdf --preset sticker.json --set bleed_mm=3
```

A preset always belongs to one job – a CutContour preset cannot be used for `separate`. Values with special characters (e.g. `font_map` with Windows paths) are easiest to enter in the preset instead of with `--set`.

Presets from the program

It is even easier in the program itself: the windows for CutContour, CMYK/crop and the print dialog have a **Preset** line at the top. Set the settings as desired, **Save…** and give a name. The command line then finds the preset by its name – without a path:

```
passermark-cli presets cutcontour
passermark-cli cutcontour bogen.pdf bogen-cut.pdf --preset "Sticker rund"
```

The presets are stored as JSON files per job – on Linux in `~/.config/passermark/presets/`, on Windows in `%APPDATA%\Passermark\presets\`. In the program: workspace **Automation** → **Presets** opens the folder. Print presets (booklet, poster, step & repeat) belong to the job `impose`.

4 · Whole folders at once

Linux (terminal): cut all PDFs from `eingang`, results to `fertig`, errors are reported.

```
mkdir -p done
for f in input/*.pdf; do
    passermark-cli cutcontour "$f" "done/$(basename "$f" .pdf)-cut.pdf" \
        --preset sticker.json --quiet || echo "ERROR: $f"
done
```

Windows (PowerShell):

```
$cli = "C:\Program Files\Passermark\passermark-cli.exe"
New-Item -ItemType Directory -Force -Path done | Out-Null
Get-ChildItem input\*.pdf | ForEach-Object {
    & $cli cutcontour $_.FullName "done\$($_.BaseName)-cut.pdf" `
        --preset sticker.json --quiet
    if ($LASTEXITCODE -ne 0) { Write-Host "ERROR: $($_.Name)" }
}
```

5 · Tips and errors

Message / problem	Solution
No motif found	Lower the tolerance (`--set detect.tolerance=15`) or set the mode (`--set detect.mode=color`)
Spots are detected as motifs	Raise the tolerance, increase `detect.min_size_mm`
Parts of a logo get separate contours	Increase `detect.gap_mm`, e.g. 3
Takes a long time	Keep `dpi` at 200; 300 easily takes twice as long. Passermark automatically uses several CPU cores for large pages with many objects
The computer should stay free for other work	Limit the cores: Linux `PASSERMARK_WORKERS=2 passermark-cli …`, Windows (PowerShell) put `$env:PASSERMARK_WORKERS=2` in front; 1 = no parallelism
Preset is for “…”	The preset belongs to a different job
Value is read as text instead of a number	Use a decimal point: `bleed_mm=2.5` (not 2,5)
Spaces in the file name	Put names in quotation marks: `"mein bogen.pdf"`

6 · Outlook

The command line grows with Passermark. The next versions add the following – each with case studies in this manual:

Version	New in the command line
1.11	Project folders (watcher): drop a file into a folder – a finished, printed or saved result comes out, one preset per folder